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Vulnerability assessment of China–Europe Railway Express multimodal transport network under cascading failures

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ABSTRACT

Transshipment and transnational long-haul transport make the multimodal transport network (MTN) vulnerable when subjected to unexpected disturbances, which may cause cascading failures. Therefore, this paper presents a dynamic layered road–rail MTN to measure the vulnerability of China–Europe Railway Express (CR Express) MTN. The statistical analysis of the topological characteristics and the final simulation results show that the layered road–rail MTN with priority connection mechanism exhibits power-law distribution characteristics. The final simulation results reveal that the road–rail MTN shows a better ability to suppress the risk cascading propagation when the transshipment coupling enhances the dynamic flexibility of the network. Although the optimal inter-layer transshipment coupling changes positively with the external disturbance and negatively with the intra-layer coupling in special cases, the moderate inter-layer coupling is still the most effective in improving the anti-cascading ability of general MTN. These findings provide recommendations for the CR Express MTN to suppress the cascading failures.

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1. Introduction

In 2013, the Chinese government proposed a national strategy, "The Belt and Road Initiative" (B&R) [1], which aims to develop a new economic partnership between Europe and Asia. The China–Europe Railway Express (CR Express) is an important part of the B&R to build the inland international trade channel, which is the international container train among China, Europe, and the countries along the Belt and Road, with fixed train numbers, routes, and schedules. Due to the shorter transportation time and the cost advantage under the government subsidy, the CR Express has developed rapidly. From the opening of the first train in 2011 until the end of March 2021, the CR Express has made 33600 trips, including 12400 made in 2020.

The emergence of the CR Express has become a catalyst for the development of inland multimodal transport, the United Nations Conference on Trade and Development (UNCTAD) defined multimodal transport as transporting goods in more than one mode of transport under a single contract [2]. In the past, the form of multimodal transport between Asia and Europe was mostly sea–railway combined transportation. The CR Express provides a new option for container trade between Asia and Europe through road–rail combined transportation. Based on the CR Express, complex transportation

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routes and the passing hub cities in the road–rail combined transportation constitute a multimodal transport network (MTN).

In the booming multimodal transport industry, the failure risk of MTN has gradually attracted academic attention [3]. For example, transportation services can be disrupted by cargo congestion caused by equipment failure or climate change-related natural disasters [4]. Moreover, transshipment among different modes of transport, long-distance transportation across the continent and limited exit ports makes the CR Express MTN more vulnerable and prone to interruption than the common transport network when subjected to external disturbances. In such cases, cargo flow in cities suffering from external disturbances may be redistributed to other cities or directly transferred by other modes of transport. Other adjacent cities may fall out of operation due to sudden increases in cargo flow, resulting in the cascading failure.

With only three main corridors and three exit ports, the CR Express MTN is more susceptible to cascading failure by the interruption of node city. On August 7, 2019, due to flooding in Turpan, Xinjiang, China, one bridge crumbled and eight railway wagons were derailed, which gave rise to cargo congestion in the entire network. Unexpected risks like this have severely affected the normal cargo transportation, which resulted in the temporary suspension of some hub nodes and cascading failure of the CR Express. Therefore, studying the vulnerability of CR Express MTN to suppress cascading failure is of considerable importance in ensuring normal network operations and improving its stability. Since traditional single-layer network structures can only portray structural relationships and cargo flow based on a single transportation mode, the intrinsic transshipment characteristics of the CR Express MTN are often ignored. In the CR Express MTN, most transport hub cities can provide individual road or rail transportation capability, but providing multimodal transport through CR Express based on highway transportation is unique to CR Express MTN. Therefore, to portray the CR Express MTN in reality, it is necessary to consider the transshipment between roads and railways.

To describe the coupling relationship between different modes of transport within the network, this study presents a dynamic layered MTN to analyze its vulnerability against external disturbances when considering transshipment among multiple modes of transport and cargo flow redistribution based on coupled map lattices (CML). The contributions of this study main contain three aspects: (1) We portray multi-functional characteristics of multi-modal transportation by the dynamic layered network structure, including the inherent transshipment characteristics of multimodal transport network (MTN) and the redistribution of cargo flow. (2) Not only the complementary among different transport networks is considered, but also the flexibility and dynamics of risk cascading failures and cargo flow redistribution among different transport networks are taken into account. (3) We validate the power-law distribution characteristics of the China–Europe Railway Express multimodal transport network (CR Express MTN), which is caused by the priority connection mechanism of the CR Express.

This study is organized as follows: the next section reviews previous studies about vulnerabilities of transport networks and cascading failure. Sections 3 and 4 describe the characteristics of CR Express MTN and introduce the CML model respectively. Then, the risk cascading propagation process is simulated and discussed in Section 5. Finally, conclusions and future research directions are provided in Section 6.

2. Literature review

2.1. Risks and vulnerabilities of transport networks

Due to the increasing complexity of the transport network structure, literature on the characteristics and vulnerability of transportation networks based on structure and function [5] has grown rapidly over the past few decades. In terms of structure, many scholars analyzed the vulnerability according to the structural characteristics of the transport network. Gao et al. [6] developed a Ricci model to measure road network vulnerability based on the topology. Tang and Huang [7] proposed a Bayesian network method for seismic vulnerability assessment of an urban road network. Besides, the vulnerability of transportation networks was also studied from the perspective of network capability and characteristics. Bell et al. [8] used capacity-weighted spectral partitioning to investigate capacity bottlenecks and transport network vulnerability. Voltes-Dorta et al. [9] analyzed the vulnerability of European air transport networks to the closure of major airports based on passenger flows.

However, for the MTN, the capacity of each node city is not only related to the cargo flow and topology but also greatly affected by the transshipment capacity of the city. The special network structure endows MTN with unique characteristics different from the general transport network, so the traditional topological network structure cannot fully describe the characteristics for transshipment through multiple modes. Hence, a layered network structure that exhibits multi-functional characteristics of multi-modal transportation is proposed to portray the MTN.

2.2. Transport network vulnerability under risk cascading propagation

Although some studies have analyzed the vulnerability of MTN, only a few studies analyzed network vulnerability by considering the risk cascading propagation. Most of the existing researches on cascading failures of transport networks are single networks [10,11], few of which discussed the interdependence of multiple transport networks [12]. Li et al. (2019) [13] examined the vulnerabilities of complementary and interdependent transport networks, but they neglected the flexibility of network functions and risk transfer between different modes of transport. Unlike most previous works,

Table 1
Comparison with existing studies on network vulnerability and cascading failures.

Paper	Vulnerabilities of transport networks		Vulnerability under cascading failures	
	Structure	Function	Transportation mode	Transport flexibility and dynamics
Tang and Huang, 2019	Single-layer topology	NA	Highway	NA
Bell et al. 2017	Single-layer topology	Capacity	Highway	NA
Voltes-Dorta et al. 2017	Single-layer topology	Passenger flow	Air transport	Passenger flow redistribution
Sun et al. 2018	Single-layer topology	Passenger flow	Subway	Load redistribution
Li et al. 2019	Multi-layer topology	Running timetable and position	Railway and air transport	NA

which only considers the interdependence among different transport networks, we take into account the flexibility and dynamics of cargo flow redistribution and risk cascading failures among different transport networks, that is achieved by the inter-layer transshipment coupling.

It is noted that in an MTN, the interaction of topology, cargo flow and transshipment make the cascading failure in MTN more special and complicated. There are several models that have been used to solve cascading failures in interdependent networks, such as the load redistribution model [12,14], percolation theory [15,16], and CML [10,17]. The synchronization and chaos of the transportation network show that it is a nonlinear dynamic system, the percolation theory and redistribution model cannot perfectly describe the spatiotemporal dynamics of MTN. However, CML is a spatiotemporal dynamic model to study the joint state changes with time and space. Thus, in this study, CML is used to characterize the cascading propagation of risks in MTN [18]. To the best of our knowledge, there is no research on the transshipment characteristics of MTN through layered network structure, which will greatly affect the redistribution of cargo flow and the cascade propagation of risk in the network. Therefore, a dynamic layered network structure with transshipment characteristics is used to analyze the impact of risk cascading propagation in MTN on network vulnerability. By comprehensively comparing the existing research on network vulnerability and cascading failures (Table 1), it can be found that the existing research has not yet characterized the multi-functional characteristics of multi-mode transport through layered network structure.

Moreover, the existing studies neglected the flexibility of network functions and risk transfer between different modes of transport. Therefore, the topological characteristics, the redistribution of cargo flow based on transshipment and the complementary of the highway network and railway network are incorporated into the analysis process simultaneously through the dynamic layered network structure.

3. Character description of MTN

L-space and P-space are two general methods for constructing transport network models [19], both of which take hub cities as nodes. However, the transshipment characteristics make the cities in the MTN connected to themselves, neither L-space nor P-space can describe this characteristic effectively. Thus, we propose a novel network construction method, ML-space, which is a multi-layered L-space model.

To better analyze the dynamic vulnerability and cascading propagation, the existing studies have described the relationship between different transport modes (multi-modal) follow the following principles: If a city has two modes of transport, then the interaction between the networks composed of these two modes of transport (railway network and highway network) can be expressed through the inter-layer links to analyze the topological characteristics and dynamic behavior of the entire system [13,20,21]. This principle has been used in many real-world systems with multi-modal characteristics, including international trade systems [22], urban rail transit systems [17,21], railway systems [23]. Therefore, the transshipment characteristics of MTN are characterized by the connection between layers. From the Land Bridge Logistics Alliance[1] and the Belt and Road Portal,[2] we obtain routes, the number of trips and freight volume of each hub city of the CR Express in 2020. From the official website of the Ministry of Transport of the People's Republic of China,[3] we have access to the road routes that are available in the hub cities of CR Express. By collecting and collating these data, a layered CR Express MTN is built based on 22 key node cities involved in China Railway Express Construction and Development Plan (2016–2020) (Fig. 1).

Since the CR Express is generally bidirectional and the cargo flows on different routes are unequal, Fig. 1 is an undirected weighted network in which cargo can be transported in both directions along the network structure. At each layer, the coupling effect occurs between the cities through the transport lines and cargo flow, which are called topological coupling and cargo-flow coupling. Specifically, the topological coupling is expressed by the network structure in each layer, the more the edges of each node city are connected to other cities, the greater the topological coupling strength. The large topological coupling of a certain node city has a positive effect on the redistribution of cargo flow, but it also speeds up the cascade propagation of risk along the network structure in each layer. The cargo-flow coupling and

1 http://www.landbridge.com/shuju/.
2 https://www.yidaiyilu.gov.cn/.
3 http://www.mot.gov.cn/shuju/.

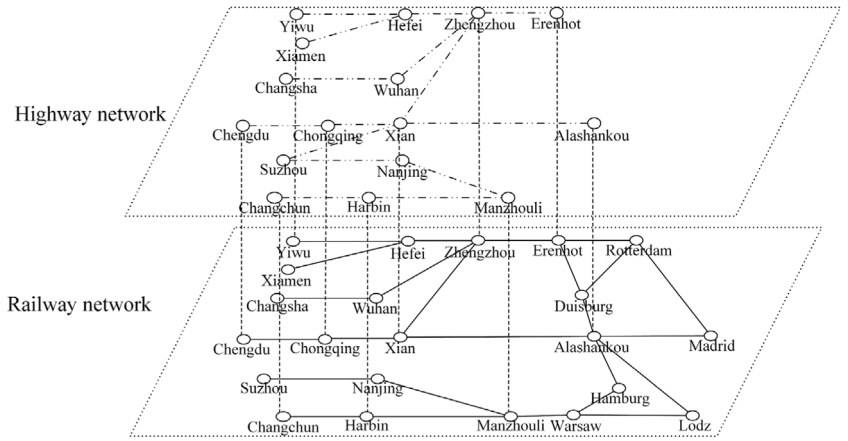
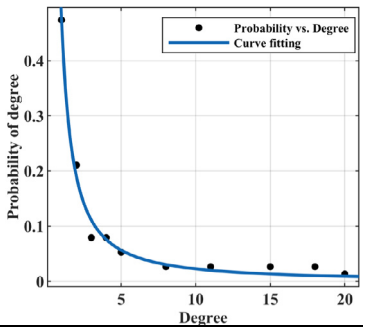


Fig. 1. Layered CR Express MTN.

Table 2
The topological characteristics of CR Express MTN.

General model						
$p(x) = ax^b$						
Coefficients (With 95% confidence bounds)		Goodness of fit				
a	b	SSE	R^2	Adjusted R^2	$RMSE$	
0.4755 (0.4389, 0.5111)	-1.326 (-1.489, -1.163)	0.002	0.989	0.988	0.016	



topological coupling share the network structure in each layer, but the cargo-flow coupling is expressed by the realistic cargo flow on the transportation route connecting two cities in the layer, that is, the weight of the edges in the intra-layer network structure. Large cargo-flow coupling indicates that the cargo flow between two cities is large, which will positively promote the cascading failure of node cities.

The transshipment coupling between layers is used to characterize the coupling effect between different transshipment modes, which is represented by the inter-layer network structure. The edges between the layers are to indicate whether a city has a transshipment capacity. If a city (e.g., Xi'an) has the ability of road–rail combined transportation, then there is an edge between the highway network and the railway network for that city. The large transshipment coupling has a positive effect on the redistribution of cargo flow in the city, but at the same time it speeds up the cascade propagation of congestion risk along the inter-layer network structure.

In view of only three key outbound ports of the CR Express, it is reasonable to suspect that the CR Express MTN is vulnerable to deliberate attacks on critical exit ports. On this basis, to better understand and analyze the structure of CR Express MTN, statistical analysis and curve fitting are carried out, the specific analysis process is as follows:

The toolbox of curve fitting in MATLAB is called, and the power-law distribution function is selected for preliminary fitting, and the analysis results in Table 2 are obtained.

The curve fitting diagram on the right of Table 1 shows the functional relationship between node degree and the probability of occurrence of node degree in the CR Express MTN. The node degree is the number of edges directly connected to the nodes in the layered network structure, including the edges between layers and within layers, which can be calculated directly through the network topology. The degree distribution $p(k)$ is the probability of nodes with a degree k among all nodes [24]. $p(k) = n(k)/\sum_{i=1}^{N} n(i)$, where $n(k)$ is the number of nodes with degree k in the network, N is the largest node degree in the network, $n(i)$ denotes the number of nodes with degree i. According to the statistics of Table 1, the sum of squares due to error (SSE) is 0.002, the root mean squared error (RMSE) is 0.016, both of which are close to 0, so the model selection and fitting effect are good. The degree-of-freedom adjusted coefficient of determination $R^2 = 0.989$, which shows that the interpretation ability of x for $p(x)$ is stronger and the fitting degree is better.

Since the curve fitting can only approximate the distribution judgment, to further verify the characteristic of the CR Express MTN, the Kolmogorov–Smirnov (K–S) goodness-of-fit test based on the parameter estimation of the maximum

Table 3
K-S analysis table.

$\hat{p}(x_i)$	0.861	0.405	0.237	0.156	0.043	0.018	0.010	0.006	0.004	0.003
$\hat{p}(x_{(i)})$	0.003	0.004	0.006	0.010	0.018	0.043	0.156	0.237	0.405	0.861
$\frac{i}{n}$	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
$\frac{i-1}{n}$	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
$\frac{i}{n} - \hat{p}(x_{(i)})$	0.097	0.196	0.294	0.390	0.482	0.557	0.544	**0.563**	0.495	0.139
$\hat{p}(x_{(i)}) - \frac{i-1}{n}$	**0.003**	−0.096	−0.194	−0.290	−0.382	−0.457	−0.444	−0.463	−0.395	−0.039

likelihood estimation (MLE) is performed. Firstly, the hypothesis H_0 is proposed: the node degree has the characteristics of the power-law distribution. And the K–S test based on the MLE is used to verify this assumption. The probability density function is $p(x) = (\lambda - 1)/x_{\min} \cdot (x/x_{\min})^{-\lambda}$, where $x \geq x_{\min}$, $(\lambda - 1)/x_{\min}$ is a standardized constant.

Then the MLE is used to estimate the parameter λ in the model. The MLE of λ can be expressed as: $\hat{\lambda} = 1 + n[\sum_{i=1}^{n} \ln(x_i/x_{\min})]^{-1}$ [25], where $i = 1, \ldots, n$. n is the number of cities with different node degree. According to the node degree, we can get $\hat{\lambda} \approx 1.861$, $\hat{p}(x) = 0.861x^{-1.861}$, and the K-S test accepts H_0 if $D = \max \left|F(x) - \hat{F}_n(x)\right| < D_{\alpha,n}$, where α is the level of significance, here we take $\alpha = 0.001$, $F(x)$ is the empirical distribution function, $D_{\alpha,n}$ is the quantile.

Let $D^+ = \max \left|i/n - \hat{p}(x)\right|$, $D^- = \max \left|\hat{p}(x) - (i-1)/n\right|$, and it turns out that $D = \max(D^+, D^-)$. So through the following table we can get $D^+ = 0.563$, $D^- = 0.003$, and then $D = \max(D^+, D^-) = 0.563$. By checking the K-S table, we find that $D_{\alpha,n} = D_{0.001,10} = 0.580 > D$. So we fail to reject the H_0 (see Table 3).

The characteristic of power-law distribution of the CR Express means a few hub cities are connected to most cities. Because there are only three exit ports for the CR Express, so these three cities must be connected to most other cities. This priority connection mechanism for connecting limited outbound ports (Alashankou, Erenhot and Manzhouli) creates the power-law distribution characteristics of the CR Express MTN, which is expressed as vulnerability to malicious disturbances and robustness to random disturbances. If malicious attacks disturb the hub city of the CR Express MTN, the existing cargo flow of this city would be redistributed to other cities, which eventually leads to cascading failure that in turn leads to variation in the network structure and vulnerability. Therefore, the analysis of the characteristics of the MTN inspires this study to examine the impact of cascading failure on the vulnerability of CR Express MTN under malicious and random attacks.

4. Cascading failure model of MTN based on CML

In MTN, cascading failure is not only related to the network topology but also the cargo flow redistribution and choice of transshipment modes. Consequently, the transshipment mode is used as one of the coupling relationships within the network, and the redistribution of cargo flow is combined to analyze the cascading failure based on CML. Here, we construct a model to depict the dynamic process of cargo transferring inter- and intra-network based on the CML model presented by Sun et al. [10]. Eq. (1) illustrates this model,

$$x_i(t+1) = \left| \begin{array}{l} (1 - \varepsilon_{topol} - \varepsilon_{flow} - \varepsilon_{trans}) f(x_i(t)) + \varepsilon_{topol} \sum_{j=1, j\neq i}^{N} k_{ij} f(x_j(t))/K_i^{\deg} \\ + \varepsilon_{flow} \sum_{j=1, j\neq i}^{N} w_{ij} f(x_j(t))/W_i^{flow} + \varepsilon_{trans} \sum_{j=1, j\neq i}^{N} v_{ij} f(x_j(t))/V_i^{trans} \end{array} \right|, \; i = 1, \ldots, N \tag{1}$$

where $x_i(t+1)$ is the state of city i that determines whether the city is still capable of providing transportation services at time $t+1$. ε_{topol} is a coefficient to measure the coupling strength of network topological structure in the same layer, representing the correlation between cities. ε_{flow} refers to another coupling coefficient used to measure the coupling strength of network cargo flow by the actual cargo flow between cities, where the large cargo flow between two cities indicates strong coupling. ε_{trans} indicates the coupling strength between different layers, measured by the transshipment modes, where higher number of transshipment modes indicates strong coupling. Because node state changes cannot be completely affected by only one coupling relationship, so ε_{topol}, ε_{flow}, ε_{trans} have constraints that ε_{topol}, ε_{flow}, $\varepsilon_{trans} \in (0, 1)$, $\varepsilon_{topol} + \varepsilon_{flow} + \varepsilon_{trans} \leq 1$.

The chaotic logistic map is expressed as $f(x) = \mu x(1-x)$, where $0 \leq x \leq 1$, $1 \leq \mu \leq 4$ [26]. Peitgen et al. [26] have demonstrated that when $0 \leq x \leq 1$, the logistic map works in a chaotic state, that is, the sequence generated by the initial condition x_0 under the action of logistic map is aperiodic and non-convergent, sequences generated outside this range must converge to a specific value. When the value of μ satisfies the condition of $3.5699456 \leq \mu \leq 4$, especially when it is closer to 4, the logistic map moves unstable within a finite iteration, and its long-term dynamic behavior will show random properties. Hence, this study represents chaotic logistic map as $f(x) = 4x(1-x)$, $0 \leq f(x) \leq 1$ if $0 \leq x \leq 1$ to express the capacity of city i.

When $0 \leq x_i \leq 1$, it means that the cargo flow in city i within the scope of its cargo handling capacity. Adjacency matrix k_{ij} indicates the connection relationship between nodes, if city i and j have a direct connection, $k_{ij} = k_{ji} = 1$,

otherwise, $k_{ij} = k_{ji} = 0$; $K_i^{\deg}$ refers to the degree of city i; w_{ij} indicates the cargo flow between city i and city j. W_i^{flow} refers to the sum of cargo flow to and from city i.[4] v_{ij} indicates the transshipment modes between city i and j. V_i^{trans} refers to the total transshipment modes of the city i in the network.[5] The absolute value symbols in the equation are used to ensure that the state of each node is non-negative.

If for all t, $0 \leq x_i(t) \leq 1$, then city i is in a stable operation state. If city i fails at time c, $x_i(c) \geq 1$, then $x_i(t) = 0, t > c$. This means that the city's cargo flow exceeds the capacity of the city due to congestion of the transportation route or the redistribution of the cargo flow, which causes the city to be out of operation. We also assume that in the absence of external disturbances, the initial states of all nodes are within the range of (0, 1), maintaining a normal operation state. At time c, the MTN is interrupted due to unexpected external disturbances, causing node i to be out of operation, which is equivalent to applying an external disturbance $R(R \geq 1)$ to node i at time c. The city interrupted by different external disturbances R has different cargo redistribution speeds, which have different influences on the state of the surrounding cities. Therefore, the larger the R, the greater the external disturbance to the city i, the faster the risk propagates along the network, the faster the state of the surrounding city changes, and the more likely the cascading failures will eventually occur. Thus,

$$x_i(c) = \left| \begin{array}{l} (1 - \varepsilon_{topol} - \varepsilon_{flow} - \varepsilon_{trans}) f(x_i(c-1)) + \varepsilon_{topol} \sum_{j=1, j\neq i}^{N} k_{ij} f(x_j(c-1))/K_i^{\deg} \\ + \varepsilon_{flow} \sum_{j=1, j\neq i}^{N} w_{ij} f(x_j(c-1))/W_i^{flow} + \varepsilon_{trans} \sum_{j=1, j\neq i}^{N} v_{ij} f(x_j(c-1))/V_i^{trans} \end{array} \right| + R \tag{2}$$

At this time, city i of MTN is out of operation owing to unexpected emergency, and then the status of MTN changes as cargo flow redistributes and transshipment. All neighbor cities of the city i are affected at time $c + 1$, and the state transition is calculated according to Eq. (1). Recalculating the node status value may be greater than 1, which triggers a new round of cascading failure.

Finally, to measure the network vulnerability, the proportion of failed cities is proposed to describe the extent to which cascading failure affects the network. It is shown as $P(t) = N'(t)/N$ $P(t) = N'(t)/N$, where $P(t)$ denotes the proportion of failed cities at time t, $N'(t)$ is the number of failed cities, and N is the number of whole cities.

5. Vulnerability analysis of CR express MTN

In this part, based on the real data from the Land Bridge Logistics Alliance, the Ministry of Transport of the People's Republic of China and the Belt and Road Portal,

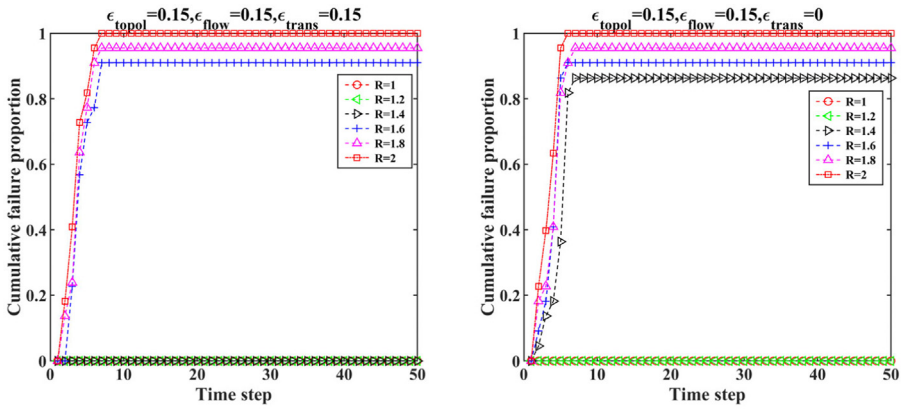
the risk cascade propagation process of CR Express MTN considering transshipment coupling is verified. Given the existence of the power-law distribution characteristic of CR Express MTN, two types of attacks, random and malicious, are taken into account for networks with and without transshipment coupling.

To make the simulation results and conclusions more credible and universal, the parameters of topological coupling, cargo-flow coupling and transshipment coupling are set according to the average coupling of cities in the realistic CR Express MTN and the existing literature. The topological coupling coefficient and cargo-flow coupling coefficient can be set as $\varepsilon_{topol} = 0.15$ and $\varepsilon_{flow} = 0.15$ for the general CR Express MTN,[6] which is also similar to the parameter settings in the existing research [10,27]. For the transshipment coupling coefficient, in order to ensure the simulation process meets the basic requirements of the CML model, that is $\varepsilon_{topol} + \varepsilon_{flow} + \varepsilon_{trans} \leq 1$, in the case where the cargo-flow coupling and topological coupling changes in the range of [0.15, 0.65], it is assumed that the transshipment coupling coefficient is also 0.15, which is consistent with the low level of multimodal transport development and the weak transshipment capacity in the CR Express system. Besides, in order to overcome the limitation of the small sample size, we take the average of 5000 simulation experiments as the final result.

[4] Based on the statistics of the number of freight trains running in each city that runs CR Express on the Land Bridge Logistics Alliance platform, the freight volume between these cities and European cities can be calculated. Then, according to the route map of the CR Express, the cities through which the CR Express passes can be identified. Thus, the freight volume of each node city on each route can be summed up, and the freight volume between each node city of CR Express MTN can be obtained.

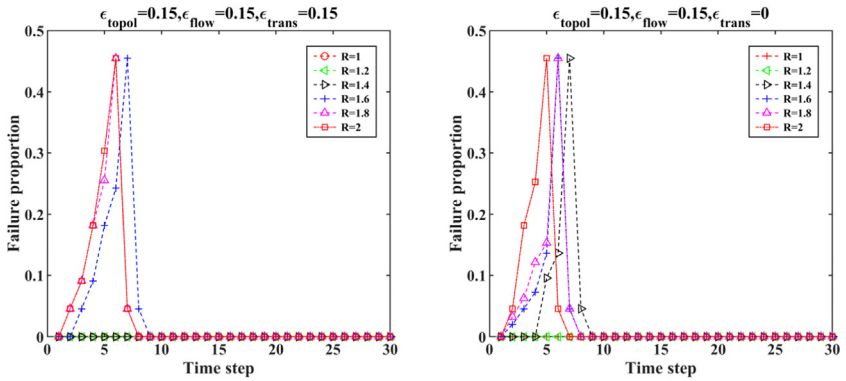
[5] As the main transportation mode of CR Express MTN is road–rail combined transportation, whether there exist multiple transshipment modes between two cities can be judged by the availability of the CR Express and highway transport in both cities simultaneously. Therefore, according to the Belt and Road Portal, a list of cities that regularly run the CR Express can be compiled, from which the transshipment modes between cities can be figured out.

[6] In the layered CR Express MTN (Fig. 1), there are 6 cities with a node degree 3 according to the statistics, and the number is maximum. To make the simulation results more general and universal, the coupling strength of a city with node degree 3 is taken as the regular topological coupling coefficient. The topological coupling coefficient of a city with node degree 3 is about 0.15. For the cargo-flow coupling coefficient, according to data from the Land Bridge Logistics Alliance platform, taking Xi'an, which ran the largest number of CR Express in 2020, as the base city with the largest cargo-flow coupling. The average cargo-flow coupling intensity of the CR Express MTN can be calculated to be 0.15, which is consistent with the above topological coupling coefficient.



(a) Considering transshipment coupling (b) Without considering transshipment coupling

Fig. 2. Cumulative failure proportion of CR Express for random attacks on different *R*.



(a) Considering transshipment coupling (b) Without considering transshipment coupling

Fig. 3. Failure proportion of CR Express for random attacks on different *R*.

5.1. Impact of external disturbance amplitude R

In this section, we study the impact of different external disturbance amplitudes on the vulnerability of the network with and without considering transshipment coupling ($\varepsilon_{trans} = 0.15$ and $\varepsilon_{trans} = 0$), so as to identify the disturbance threshold of cascading failure in the MTN. Two ways of disturbance are exerted, one is to randomly select nodes to exert disturbances and the other is to select the node with the largest degree of importance for malicious attacks.

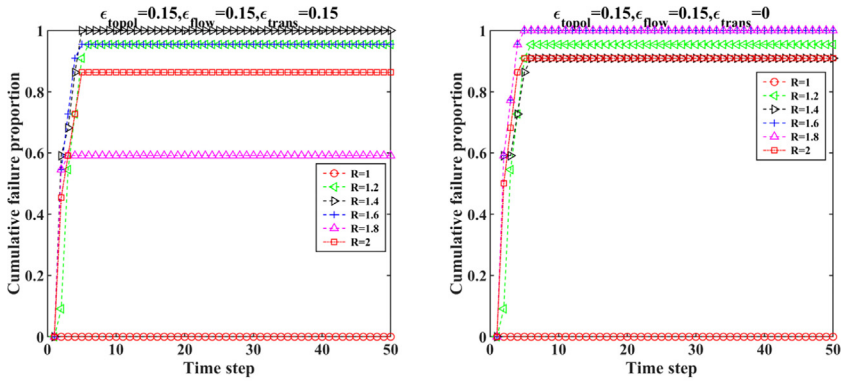
5.1.1. Randomly selected nodes are subject to external disturbances

Under random attacks, the threshold of CR Express MTN is analyzed with different disturbance intensities. Given $\varepsilon_{topol} = \varepsilon_{flow} = 0.15$, disturbance intensity varies from 1 to 2. As can be seen from Fig. 2, considering transshipment coupling leads to a higher threshold of external disturbance ($R \geq 1.6$) than not considering transshipment coupling ($R \geq 1.4$, Fig. 2(b)). Fig. 3 also shows that as external disturbance increases, the cascading failure occurs and the peak is reached earlier. According to the threshold analysis of cascading failure, a large external disturbance results in the early occurrence of risk cascading propagation. Therefore, considering the transshipment coupling enables the MTN to better suppress the risk cascading propagation under random attacks, which encourages the adoption of multimodal transport in international long-haul transport across Eurasia.

5.1.2. The node with the largest degree of importance is subject to malicious attack

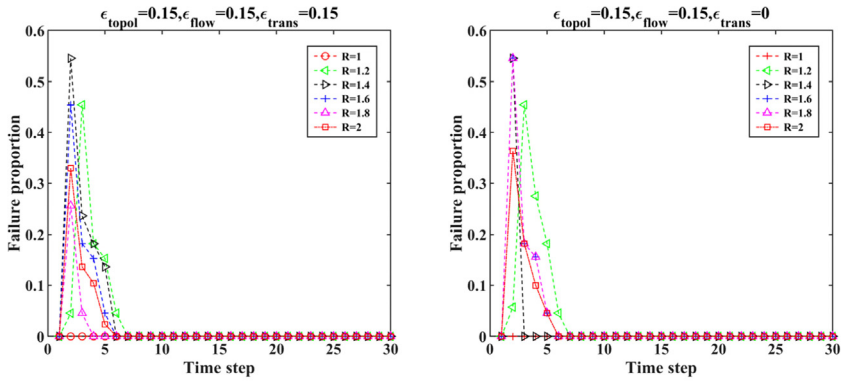
In this case, the most important hub node is selected for malicious attack, namely, Alashankou, which is one of the three exit ports for the CR Express. Given $\varepsilon_{topol} = \varepsilon_{flow} = 0.15$, disturbance intensity varies from 1 to 2 when Alashankou is subject to malicious attack, as shown in Fig. 4.

By comparing Figs. 2 and 4, it can be found that cascading failures occur earlier and more strongly under malicious attacks than under random attacks. Moreover, cascading failure occurs faster and causes greater damage to malicious



(a) Considering transshipment coupling (b) Without considering transshipment coupling

Fig. 4. Cumulative failure proportion of CR Express for malicious attack on different R.



(a) Considering transshipment coupling (b) Without considering transshipment coupling

Fig. 5. Failure proportion of CR Express for malicious attack on different R.

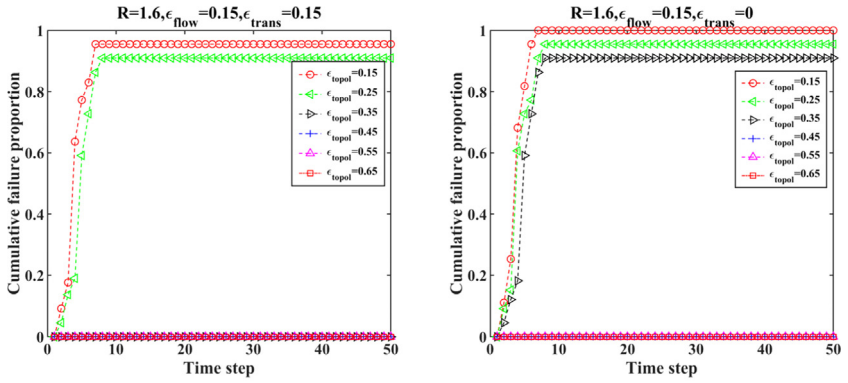
attacks compared with random attacks, which shows that CR Express MTN is more vulnerable to malicious attacks (see Fig. 5). Therefore, in order to prevent cascading failures, the redundancy of the network should be increased, especially the three exit ports (Alashankou, Erenhot, and Manzhouli). For malicious attacks, redundancy can not only provide alternatives to minimize the impact of cascading failures, but also promote network recovery by improving network resilience [28]. For the CR Express MTN, the redundancy of the network can be improved by adding exit ports and transshipment modes. For example, the opening of the Horgos port can increase the redundancy of Alashankou and alleviate the cascading failure when the port is attacked. Similarly, the opening of Suifenhe port in September 2020 has further eased pressure on exports of cargos from Manzhouli and Erenhot.

5.2. Analysis of topological coupling coefficient ε_{topol}

This part analyzes the vulnerability of CR Express MTN under different topological coupling coefficient ε_{topol}. Given $\varepsilon_{flow} = 0.15$, $R = 1.6$, $\varepsilon_{trans} = 0.15$ when considering the transshipment coupling, $\varepsilon_{trans} = 0$ when not considering the transshipment coupling, we select specific and random nodes to exert perturbations in the network.

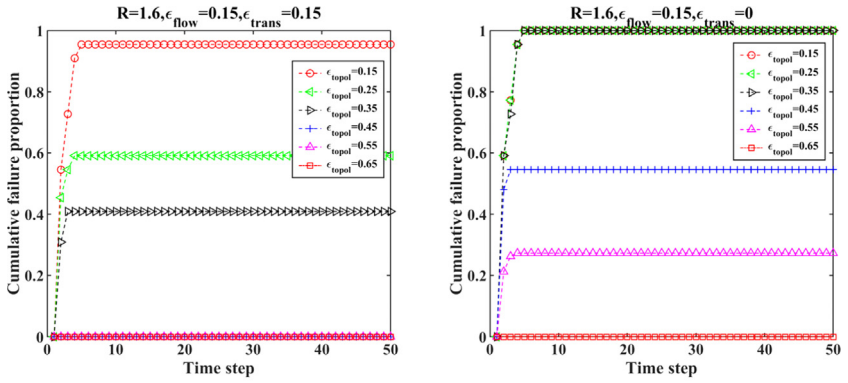
5.2.1. Randomly selected nodes are subject to external disturbances

As can be seen from Fig. 6, the smaller the strength of topological coupling, the more likely the risk cascading propagation will occur in the network when a randomly selected node is attacked. Besides, by comparing Figs. 6(a) and (b), it can be found that the cascading failure caused by random disturbance can be suppressed by considering the transshipment coupling. In Fig. 6(a), the cascading failure does not take place until $\varepsilon_{topol} < 0.35$, which shows that high topological coupling can mitigate cascading failure. Although topological coupling promotes cascading risk propagation along the network, topology also promotes the redistribution of cargo flow and positive impact is greater than the negative impact. We can conclude that high coupling strength within the layer can significantly improve the anti-cascading ability of the layered MTN. Therefore, adjusting the network structure and increasing the topological coupling in MTN can



(a) Considering transshipment coupling (b) Without considering transshipment coupling

Fig. 6. Cumulative failure proportion of CR Express for random attack on different ε_{topol}.



(a) Considering transshipment coupling (b) Without considering transshipment coupling

Fig. 7. Cumulative failure proportion of CR Express for malicious attack on different ε_{topol}.

suppress cascading failure and reduce network vulnerability. For the CR Express MTN, opening new freight routes in addition to the existing three corridors and establishing other exit ports to create a more reasonable network structure can effectively reduce network vulnerability.

5.2.2. The node with the largest degree is subject to malicious attack

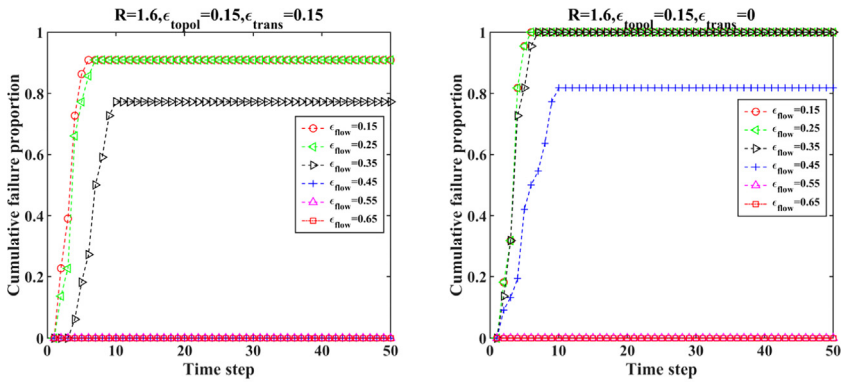
By comparing Figs. 6 and 7, it can be seen that the CR Express MTN is more susceptible to cascading failures, even if the topological coupling strength is high (e.g., $\varepsilon_{topol} = 0.55$), under malicious disturbance than under random disturbance.

5.3. Analysis of cargo-flow coupling coefficient ε_{flow}

In the following part, the vulnerability of CR Express MTN under different cargo-flow coupling coefficient ε_{flow} is studied. An external disturbance $R = 1.6$ is assumed. Given $\varepsilon_{topol} = 0.15$, $\varepsilon_{trans} = 0.15$ when considering the transshipment coupling, $\varepsilon_{trans} = 0$ when not considering the transshipment coupling, the MTN vulnerability and the threshold of cascading failure as the coupling coefficient of cargo flow changes under random and malicious attacks are identified.

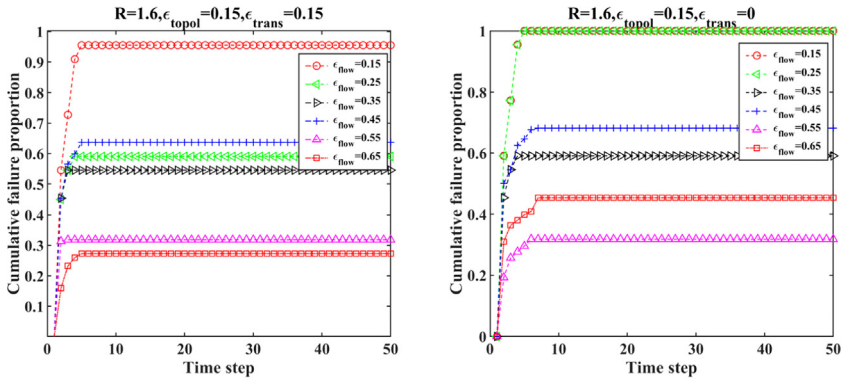
5.3.1. Randomly selected nodes are subject to external disturbances

Under a random attack, when considering the transshipment coupling, cascading failure occurs if $\varepsilon_{flow} < 0.45$ (Fig. 8(a)). Without considering the transshipment coupling, cascading failure occurs if $\varepsilon_{flow} < 0.55$ (Fig. 8(b)), which is greater. However, in both cases, the thresholds of the coupling coefficient for triggering cascading failure are greater than the coupling coefficients of the topological structure (Fig. 6), indicating that the suppression of cascading failure by cargo-flow coupling is less than the suppression of cascading failure by topological structure.



(a) Considering transshipment coupling (b) Without considering transshipment coupling

Fig. 8. Cumulative failure proportion of CR Express for random attack on different ε_{flow}.



(a) Considering transshipment coupling (b) Without considering transshipment coupling

Fig. 9. Cumulative failure proportion of CR Express for malicious attack on different ε_{flow}.

5.3.2. The node with the largest degree is subject to malicious attack

In the face of the malicious attack, the change of cargo-flow coupling cannot restrain the cascading failure effectively, even if the transshipment coupling is considered (Fig. 9(a)). In addition, the cascading failures are not only not suppressed but deteriorated to a certain extent (e.g., $\varepsilon_{flow} = 0.25$, $\varepsilon_{flow} = 0.65$). Therefore, the adjustment of inter-city cargo flow cannot effectively prevent the further propagation of risk in the network, even if there are many transshipment modes. Hence, in CR Express MTN, channeling and regulating the cargo flow is not an effective way to mitigate the risk cascading propagation.

5.4. Analysis of transshipment coupling coefficient ε_{trans}

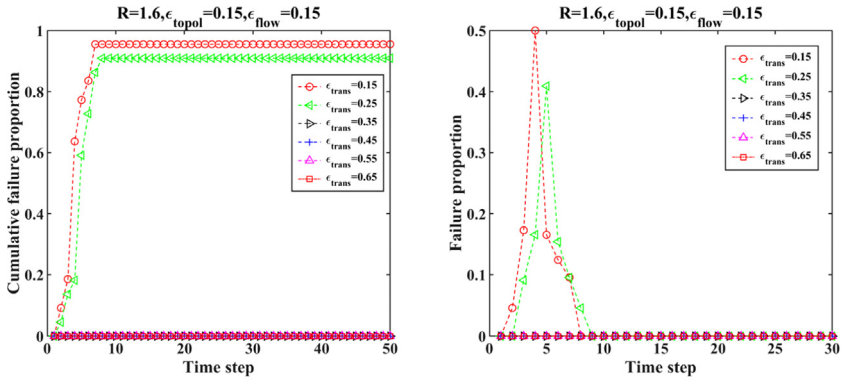
This part examines how the vulnerability of CR Express MTN varies with the transshipment coupling coefficient ε_{trans} under random and malicious attacks.

5.4.1. Randomly selected nodes are subject to external disturbances

Under random attack, in light of Fig. 10(a), the cascading failure is triggered when $\varepsilon_{trans} < 0.35$, which is similar to the analysis of the topological coupling coefficient ε_{topol} under random attacks. As the transshipment coupling coefficient increases, the risk cascading propagation in the MTN is suppressed (Fig. 10(b)) because the possible selection of more transshipment modes decreases the likelihood of cargo congestion.

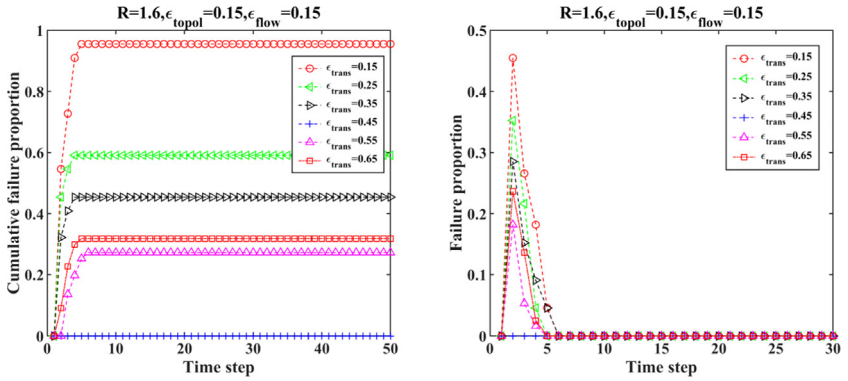
5.4.2. The node with the largest degree is subject to malicious attack

Fig. 11 shows that when $\varepsilon_{trans} < 0.45$, the network becomes increasingly vulnerable as ε_{trans} decreases. However, when $\varepsilon_{trans} \geq 0.45$, the network becomes increasingly robust as ε_{trans} decreases. This finding shows that transshipment coupling can suppress cascading failure within a certain range but exceeding a certain range promotes risk cascading propagation and increases network vulnerability. In the CR Express MTN, a smaller transshipment coupling will limit



(a) Cumulative failure proportion (b) Failure proportion of CR Express

Fig. 10. Randomly selected nodes for attack on different ε_{trans}.



(a) Cumulative failure proportion (b) Failure proportion of CR Express

Fig. 11. The node with the largest degree of malicious attack on different ε_{trans}.

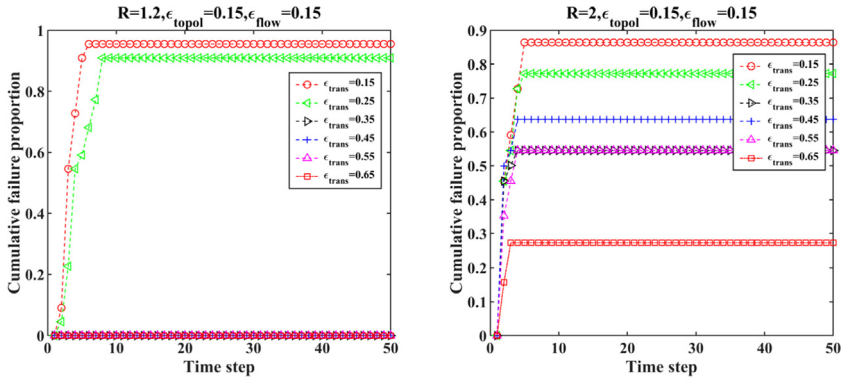
the redistribution of cargo flow, but a larger transshipment coupling will facilitate the propagation of risks along the network structure, so there is an optimal transshipment coupling. Therefore, for multimodal transport in the CR Express, the moderate transshipment coupling (e.g., under current conditions, $\varepsilon_{trans} = 0.45$) is the most advantageous to restrain the risk cascading propagation.

5.4.3. The validity analysis of model and optimal inter-layer coupling

By studying the effect of transshipment coupling on network vulnerability, we find that there is an optimal coupling strength between layers that will make the network the most capable of suppressing cascading failure, but this conclusion is based on the assumption of fixed external disturbance R and regular intra-layer coupling coefficient ε_{topol} and ε_{flow}. Although the inter-layer coupling coefficient is set according to the regular situation in the reality of the CR Express MTN, however, it does not fully reflect the complex external conditions and the special coupling characteristics of some cities. Therefore, the optimal inter-layer coupling is further verified by the variation of external disturbance, topological structure and cargo-flow coupling in special cases, which further improves the rationality of the model and the reliability of the conclusions.

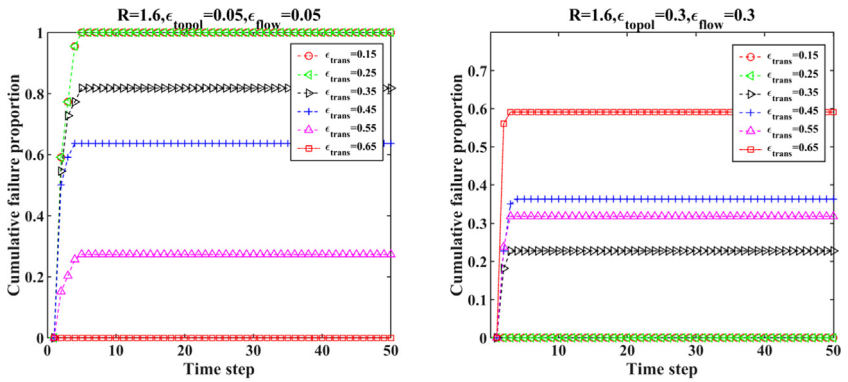
In order to be close to the actual operation of CR Express MTN, the coupling coefficients in special cases are set according to the data of the Land Bridge Logistics Alliance platform. Taking Harbin and Zhengzhou, which have small and large cargo flow, as examples, Harbin and Zhengzhou ran CR Express 213 times and 1106 times respectively in 2020. Based on Xi'an, which ran the largest number of CR Express in 2020, the smaller cargo-flow coupling coefficient can be expressed as $\varepsilon_{flow} = 213/3670 \approx 0.05$, and the larger cargo-flow coupling coefficient can be expressed as $\varepsilon_{flow} = 1106/3670 \approx 0.30$. Similarly, the smaller and larger topological coupling coefficients can also be approximately expressed as $\varepsilon_{topol} = 0.05$, $\varepsilon_{topol} = 0.3$. For the intensity of external disturbance, we take $R = 1.2$ and $R = 2$ as the two special cases of small disturbance and large disturbance.

Fig. 12 can be obtained by analyzing the weak external disturbance ($R = 1.2$) and the extreme external disturbance ($R = 2$). As can be seen from Fig. 12, the weak external disturbance reduces the optimal inter-layer coupling strength



(a) Smaller external disturbance ($R = 1.2$) (b) Larger external disturbance ($R = 2$)

Fig. 12. Cumulative failure proportion of CR Express for random attacks on different R.



(a) Smaller coupling coefficient ($\varepsilon_{topol} = \varepsilon_{flow} = 0.05$)

(b) Larger coupling coefficient ($\varepsilon_{topol} = \varepsilon_{flow} = 0.3$)

Fig. 13. Cumulative failure proportion of CR Express for random attack on different ε_{trans}.

($\varepsilon^*_{trans} = 0.35$) in Fig. 12(a), the extreme external disturbance enables optimal inter-layer coupling larger (in Fig. 12(b)). This indicates that moderate inter-layer coupling can effectively suppress the risk cascading propagation only for general external disturbances, such as daily congestion, cargo delay, etc. In the face of the special external disturbance of the MTN, only moderate inter-layer coupling cannot effectively suppress risk cascading propagation, at which time the inter-layer coupling strength needs to be enhanced.

By analyzing the lower intra-layer coupling ($\varepsilon_{topol} = \varepsilon_{flow} = 0.05$) and the higher intra-layer coupling ($\varepsilon_{topol} = \varepsilon_{flow} = 0.3$), Fig. 13 is obtained. Compared to Fig. 11(a), it can be seen that higher intra-layer coupling decreases the optimal inter-layer coupling ($\varepsilon^*_{trans} = 0.25$), while lower intra-layer coupling increases the optimal inter-layer coupling ($\varepsilon^*_{trans} = 0.65$). Therefore, for a regular MTN ($\varepsilon_{topol} = \varepsilon_{flow} = 0.15$), the moderate inter-layer coupling can effectively enhance the network's anti-cascading. For a special MTN with large or small intra-layer coupling ($\varepsilon_{topol} = \varepsilon_{flow} = 0.05$ or $\varepsilon_{topol} = \varepsilon_{flow} = 0.3$), it is necessary to dynamically adjust the optimal inter-layer coupling in the opposite direction according to the intra-layer coupling, so as to achieve the balance between intra-layer coupling and inter-layer coupling.

6. Conclusions and implications

In this study, a dynamic layered MTN is established to measure the vulnerability of CR Express MTN by considering the topological coupling, cargo-flow coupling, and transshipment coupling. The intrinsic transshipment characteristics of MTN are characterized by layering the network and considering transshipment coupling. The findings are summarized as follows:

The layered road–rail MTN with priority connection mechanism exhibits power-law distribution characteristics, which means it is vulnerable to malicious disturbances. Statistical analysis and final simulation results confirm that the CR Express MTN is vulnerable to malicious disturbances, which is due to the priority connection mechanism between the cities that newly opened CR Express and the limited exit ports.

The inter-layer transshipment coupling can effectively suppress the risk cascading propagation by enhancing the flexibility of cargo flow redistribution. Although the topological coupling has the greatest impact on the risk cascading propagation, the existence of transshipment coupling can also effectively suppress the cascading propagation of risks. Many existing studies on the vulnerability and robustness of complex networks have proved that the network topology can affect the cascading propagation of risk [29,30]. However, they ignored the flexibility and dynamics of cargo flow redistribution brought by the inherent transshipment characteristics of MTN. This conclusion can be used as a reference for those non-transshipment capacity cities which have geographical advantages but are still hesitant to open the CR Express. That is, increasing transshipment coupling (opening CR Express) can not only open trade channels with Europe but also enhance their anti-cascading.

Moreover, for a general MTN, moderate inter-layer transshipment coupling can effectively enhance the anti-cascading capacity of the network. However, the optimal inter-layer coupling is not always constant. In special cases, it will change positively with the external disturbance and negatively with the intra-layer coupling. For example, for multimodal transport systems, which are often faced with sudden incidents such as freight surges, the transshipment capacity needs to be further enhanced on the basis of moderate inter-layer coupling. For multimodal transport systems with highly developed highway networks (high topological coupling), the transshipment capacity can be reduced on the basis of moderate inter-layer coupling.

The conclusions of this paper provide some implications for the CR Express MTN to suppress the cascading propagation of risks, establish an effective and robust network structure, and carry out effective risk management. In the CR Express MTN, the planners of CR Express should increase the redundancy of the transportation network according to the power-law distribution characteristic caused by the priority connection mechanism, such as the increase of exit ports (e.g., Suifenhe and Khorgas) and the planning of new corridors. Moreover, the optimal inter-layer coupling can optimize the resource deployment of CR Express operators and policy-makers, providing a reference for the development of dynamic security assessment and emergency management strategies in the CR Express MTN. For the policy-makers of the entire CR Express MTN, when the emergency resources are limited, the findings of this study can help decision-makers to make emergency resource allocation to prevent further failure propagation, the resource allocation strategy is as follows: Firstly, some emergency resources are prioritized to the failed nodes. When there are no failed nodes, some emergency resources are evenly allocated to nodes with congestion tendency (port nodes such as Alashakou, Erenhot and Manzhouli). The allocation of resources to nodes that do not have a transshipment capability is then considered, followed by other generic nodes. For railway operators, the design of transit flexibility in cities can be based on the following principles: (1) Cities in highly developed road network hubs reduce transshipments; (2) Cities vulnerable to the risks of climate change-related natural disasters or congestion increase the number of transshipments to enhance flexibility; (3) General cities provide normal road–rail combined transport to ensure optimal anti-cascading capabilities.

This study unfolds some avenues for future research. This study considers the topological structure, cargo flow, and transshipment coupling methods, but the interaction between these three may need more exploration. Moreover, the MTN has the complexity of optimal route selection when cargo flow is redistributed [31,32]. As the transshipment of multiple modes of transportation is involved, the problem of choosing the optimal distribution route will be faced when the cargo flow is redistributed. Besides, the MTN has the characteristic of low-cost–high-efficiency. The MTN uses containers as the carrier, which has advantages in transportation cost and time compared to roads in medium and long-distance transportation, but the conclusions of this study only achieve a trade-off between transportation efficiency and cascade robustness. Based on the low-cost–high-efficiency characteristic of MTN, achieving the trade-off between cost, transport efficiency and cascade robustness is a research direction worth exploring in the future.

CRediT authorship contribution statement

Bi Wang: Methodology, Software, Formal analysis, Data curation, Writing - original draft, Visualization. **Qin Su:** Conceptualization, Writing - review & editing, Supervision, Project administration, Funding acquisition. **Kwai Sang Chin:** Investigation, Writing - review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

[1] Z.Z. Shao, Z.J. Ma, J.B. Sheu, H.O. Gao, Evaluation of large-scale transnational high-speed railway construction priority in the belt and road region, Transp. Res. Part E Logist. Transp. Rev. 117 (2018) 40–57, https://doi.org/10.1016/j.tre.2017.07.007.
[2] UNCTAD, 50 years of review of maritime transport, 1968-2018: Reflecting on the past, exploring the future, 2018.
[3] X. Yang, X. (Jeff) Ban, J. Mitchell, Modeling multimodal transportation network emergency evacuation considering evacuees' cooperative behavior, Transp. Res. Part A Policy Pract. 114 (2018) 380–397, https://doi.org/10.1016/j.tra.2018.01.037.
[4] O. Cats, E. Jenelius, Beyond a complete failure: the impact of partial capacity degradation on public transport network vulnerability, Transp. B. 6 (2018) 77–96, https://doi.org/10.1080/21680566.2016.1267596.
[5] W.B. Du, X.L. Zhou, O. Lordan, Z. Wang, C. Zhao, Y.B. Zhu, Analysis of the Chinese airline network as multi-layer networks, Transp. Res. Part E Logist. Transp. Rev. 89 (2016) 108–116, https://doi.org/10.1016/j.tre.2016.03.009.
[6] L. Gao, X. Liu, Y. Liu, P. Wang, M. Deng, Q. Zhu, H. Li, Measuring road network topology vulnerability by Ricci curvature, Phys. A Stat. Mech. Appl. 527 (2019) 121071, https://doi.org/10.1016/j.physa.2019.121071.
[7] Y. Tang, S. Huang, Assessing seismic vulnerability of urban road networks by a Bayesian network approach, Transp. Res. Part D Transp. Environ. 77 (2019) 390–402, https://doi.org/10.1016/j.trd.2019.02.003.
[8] M.G.H. Bell, F. Kurauchi, S. Perera, W. Wong, Investigating transport network vulnerability by capacity weighted spectral analysis, Transp. Res. Part B Methodol. 99 (2017) 251–266, https://doi.org/10.1016/j.trb.2017.03.002.
[9] A. Voltes-Dorta, H. Rodríguez-Déniz, P. Suau-Sanchez, Vulnerability of the European air transport network to major airport closures from the perspective of passenger delays: Ranking the most critical airports, Transp. Res. Part A Policy Pract. 96 (2017) 119–145, https://doi.org/10.1016/j.tra.2016.12.009.
[10] L. Sun, Y. Huang, Y. Chen, L. Yao, Vulnerability assessment of urban rail transit based on multi-static weighted method in Beijing, China, Transp. Res. Part A Policy Pract. 108 (2018) 12–24, https://doi.org/10.1016/j.tra.2017.12.008.
[11] M. Steadieseifi, N.P. Dellaert, W. Nuijten, T. Van Woensel, R. Raoufi, Multimodal freight transportation planning: A literature review, European J. Oper. Res. 233 (2014) 1–15, https://doi.org/10.1016/j.ejor.2013.06.055.
[12] L. Tang, K. Jing, J. He, H.E. Stanley, Complex interdependent supply chain networks: Cascading failure and robustness, Phys. A Stat. Mech. Appl. 443 (2016) 58–69, https://doi.org/10.1016/j.physa.2015.09.082.
[13] T. Li, L. Rong, K. Yan, Vulnerability analysis and critical area identification of public transport system: A case of high-speed rail and air transport coupling system in China, Transp. Res. Part A Policy Pract. 127 (2019) 55–70, https://doi.org/10.1016/j.tra.2019.07.008.
[14] Y. Zhang, N. Yang, Vulnerability analysis of interdependent R & D networks under risk cascading propagation, Phys. A Stat. Mech. Appl. 505 (2018) 1056–1068, https://doi.org/10.1016/j.physa.2018.04.063.
[15] S. Havlin, N.A.M. Araújo, S.V. Buldyrev, C.S. Dias, R. Parshani, G. Paul, H.E. Stanley, Catastrophic cascade of failures in interdependent networks, Proc. Int. Sch. Phys. Enrico Fermi. 176 (2012) 311–324, https://doi.org/10.3254/978-1-61499-071-0-311.
[16] F. Wang, L. Tian, R. Du, G. Dong, The robustness of interdependent weighted networks, Phys. A Stat. Mech. Appl. 508 (2018) 675–680, https://doi.org/10.1016/j.physa.2018.05.110.
[17] F. Ma, F. Liu, K.F. Yuen, P. Lai, Q. Sun, X. Li, Cascading failures and vulnerability evolution in bus–metro complex bilayer networks under rainstorm weather conditions, Int. J. Environ. Res. Publ. Health 16 (2019) 329, https://doi.org/10.3390/ijerph16030329.
[18] F. Ren, T. Zhao, H. Wang, Risk and resilience analysis of complex network systems considering cascading failure and recovery strategy based on coupled map lattices, Math. Probl. Eng. 2015 (2015) https://doi.org/10.1155/2015/761818.
[19] V. Quera, A. Solanas, L. Salafranca, F.S. Beltran, S. Herrando, P-SPACE: A program for simulating spatial behavior in small groups, Behav. Res. Methods Instrum. Comput. 32 (2000) 191–196, https://doi.org/10.3758/BF03200801.
[20] M. Ouyang, Z. Pan, L. Hong, Y. He, Vulnerability analysis of complementary transportation systems with applications to railway and airline systems in China, Reliab. Eng. Syst. Saf. 142 (2015) 248–257, https://doi.org/10.1016/j.ress.2015.05.013.
[21] L. Hong, Y. Yan, M. Ouyang, H. Tian, X. He, Vulnerability effects of passengers' intermodal transfer distance preference and subway expansion on complementary urban public transportation systems, Reliab. Eng. Syst. Saf. 158 (2017) 58–72, https://doi.org/10.1016/j.ress.2016.10.001.
[22] M. Barigozzi, G. Fagiolo, D. Garlaschelli, Multinetwork of international trade: A commodity-specific analysis, Phys. Rev. E 81 (2010) 46104, https://doi.org/10.1103/PhysRevE.81.046104.
[23] M. Kurant, P. Thiran, Layered complex networks, Phys. Rev. Lett. 96 (2006) 138701, https://doi.org/10.1103/PhysRevLett.96.138701.
[24] S.H. Strogatz, Exploring complex networks, Nature 410 (2001) 268–276, https://doi.org/10.1038/35065725.
[25] A. Clauset, C.R. Shalizi, M.E.J. Newman, Power-law distributions in empirical data, SIAM Rev. 51 (2009) 661–703, https://doi.org/10.1137/070710111.
[26] J. Abram, H.O. Peitgen, H. Jurgens, D. Saupe, Chaos and Fractals (New Frontiers of Science), Springer Science & Business Media, 1995, https://doi.org/10.2307/3620111.
[27] K.J. Xu, C. Hong, X.H. Zhang, Q.H. Sun, N. He, M.M. Xiao, Cascades in coupled map lattices with heterogeneous distribution of perturbations, Phys. A Stat. Mech. Appl. 547 (2020) 123839, https://doi.org/10.1016/j.physa.2019.123839.
[28] X. Xu, A. Chen, S. Jansuwan, C. Yang, S. Ryu, Transportation network redundancy: Complementary measures and computational methods, Transp. Res. Part B Methodol. 114 (2018) 68–85, https://doi.org/10.1016/j.trb.2018.05.014.
[29] L. Dueñas Osorio, S.M. Vemuru, Cascading failures in complex infrastructure systems, Struct. Saf. 31 (2009) 157–167, https://doi.org/10.1016/j.strusafe.2008.06.007.
[30] O. Cats, The robustness value of public transport development plans, J. Transp. Geogr. 51 (2016) 236–246, https://doi.org/10.1016/j.jtrangeo.2016.01.011.
[31] S. Fazayeli, A. Eydi, I.N. Kamalabadi, Location-routing problem in multimodal transportation network with time windows and fuzzy demands: Presenting a two-part genetic algorithm, Comput. Ind. Eng. 119 (2018) 233–246, https://doi.org/10.1016/j.cie.2018.03.041.
[32] A. Idri, M. Oukarfi, A. Boulmakoul, K. Zeitouni, A. Masri, A distributed approach for shortest path algorithm in dynamic multimodal transportation networks, Transp. Res. Procedia 27 (2017) 294–300, https://doi.org/10.1016/j.trpro.2017.12.094.